Omni-Fusion Hackathon Video Script

Target Duration: 1.3 Minutes (~80 seconds)

[0:00 - 0:15] The Hook & Problem

Visual: High-energy montage. Fast cuts of doctors looking stressed in a busy ER. Text on screen: 'Fragmented Data.' 'Algorithmic Bias.' 'Missing Records.'

Audio: *"In high-volume emergency rooms, doctors are forced to make life-or-death cardiovascular decisions using fragmented data. Worse, existing AI models often suffer from bias and fail completely when patient records are missing."*

[0:15 - 0:35] The Multimodal Solution

Visual: Omni-Fusion logo animated. 3D glowing streams of data (ECG waves, Vitals, Text) merging into one glowing core. Text on screen: '12-Lead ECG + Vitals + History'

Audio: *"Meet Omni-Fusion - a multimodal AI platform that fuses real-time ECGs, clinical vitals, and longitudinal history to predict cardiovascular risk with unprecedented accuracy."*

[0:35 - 0:50] Resilience (Key Innovation 1)

Visual: Screen recording of the Omni-Fusion UI. A user uploads data but skips the 'historical data' section. The AI still successfully processes it. Text on screen: '100% Uptime via Dynamic Routing.'

Audio: *"What if data is missing? Our first breakthrough is Resilience. Using dynamic routing, if a patient lacks historical records, Omni-Fusion seamlessly falls back to available streams, ensuring 100% uptime in low-resource settings."*

[0:50 - 1:10] Algorithmic Equity (Key Innovation 2)

Visual: Split screen showing two diverse patients. A graph shows the 'Error Divergence' shrinking from 12.5% to <1.5%.

Audio: *"Our second breakthrough is Equity. We engineered a custom Equity-Aware Loss Function that actively penalizes the network for bias during training. We reduced the demographic error gap from 12.5% to under 1.5%."*

[1:10 - 1:30] Trust & Impact (Key Innovation 3)

Visual: Close-up of the UI showing the 1D Grad-CAM heatmap highlighting a specific T-wave on the ECG, and a SHAP waterfall chart. Doctor smiling, looking at the tablet.

Audio: *"And finally, Trust. Doctors need to understand the AI. Omni-Fusion provides full explainability - using Grad-CAM to highlight exact anomalies on the ECG, and SHAP charts to explain which vitals drove the risk score. Omni-Fusion: Equitable, explainable, and resilient cardiovascular AI."*